DEM 1110
Introduction to Demography

Social, Economic and Demographic Statistics

SARDAR CONRAD

Outline….

- Introduction
- The course: aim, classes and assignments
- Expectations
  - Test/Assignment
  - Presentation
  - Tutorials
  - Exam

- Introduction

Studies

Fun/Music/Dance

Aim…

– Nature and scope of Social, Economic and Demographic statistics
– Application of Social, Economic and Demo stats
– Concepts and Definition
– Types and Sources of Social and Economic Statistics in Demography
– Social and Economic Indicators
– Presentation of social, Economic and Demographic statistics
– Introduction to Evaluation and Adjustment of Demographic data

Nature and Scope of Social, Economic and Demographic Statistics

" Poor data hurts African countries' ability to make good policy decisions"

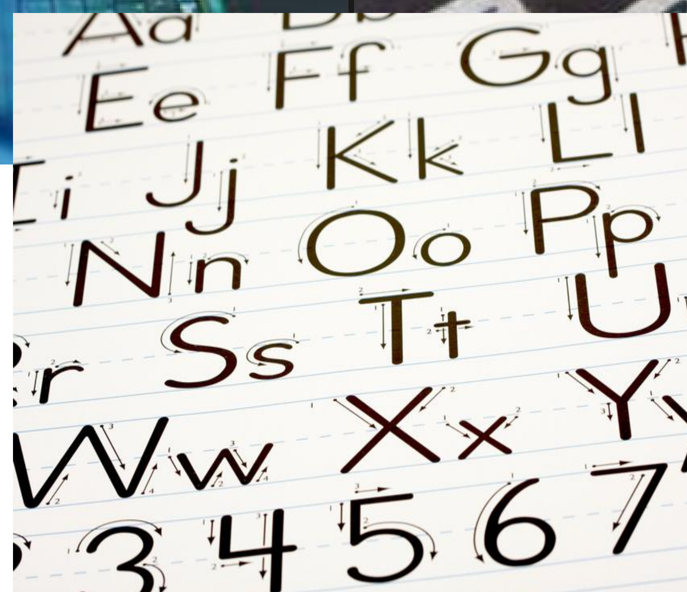
QUARTZ AFRICA..

Nature and Scope of Social, Economic and Demographic Statistics

- Data, and especially data of **good quality**, are essential for national governments and institutions to **accurately plan**, **fund** and **evaluate development activities**
- We need data/Information to:
  - Measure
  - Compare
  - Count events/phenomena/ anything

In Demography data and information are the central focus

What is Data?



What is Data?

- Data is a **collection** of numbers, symbols, sounds, actions, gestures pertaining to a phenomenon. Data may not tell a story until it is processed or analyzed.
- **Data are a representation of reality** that allows one to **perform calculations** meant to reveal **information** about the reality
- Data are the measures of properties or processes
- Data comes in **different units** of measurement

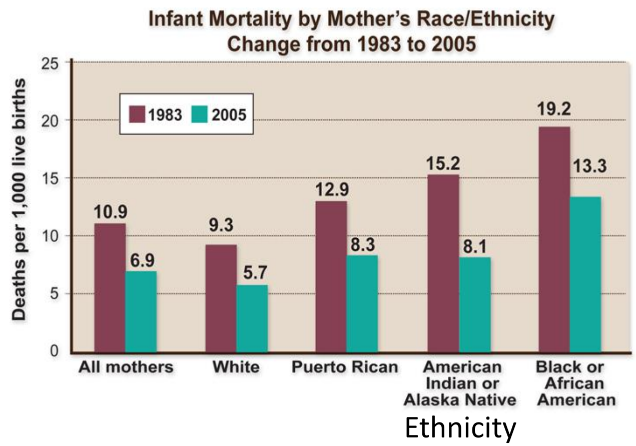
What is Data?

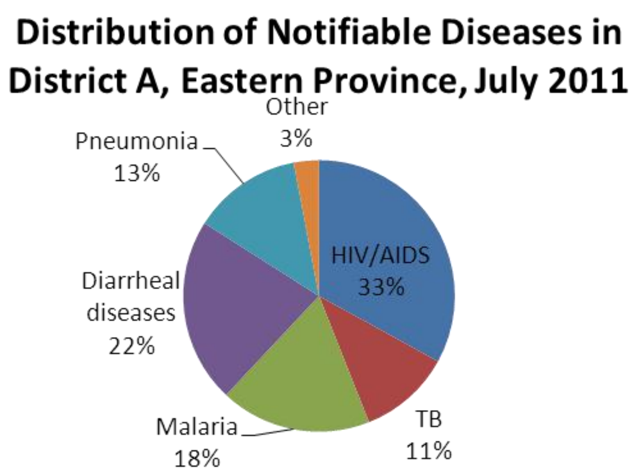
- Data is essential in understanding how things happen, change or how the program is changing.

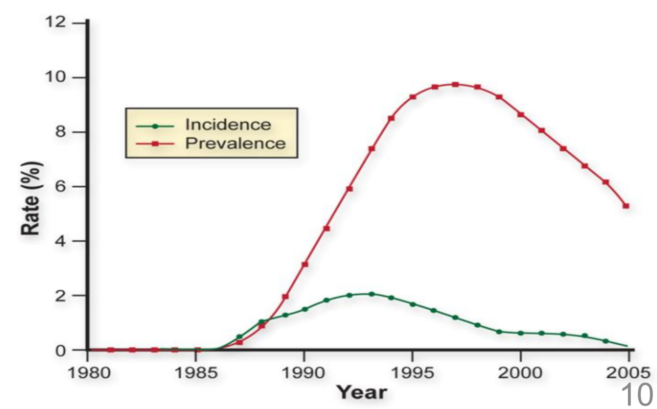
- Data are the basis from which **information is generated**, **knowledge is acquired** and **wisdom is anchored**

What is Information?

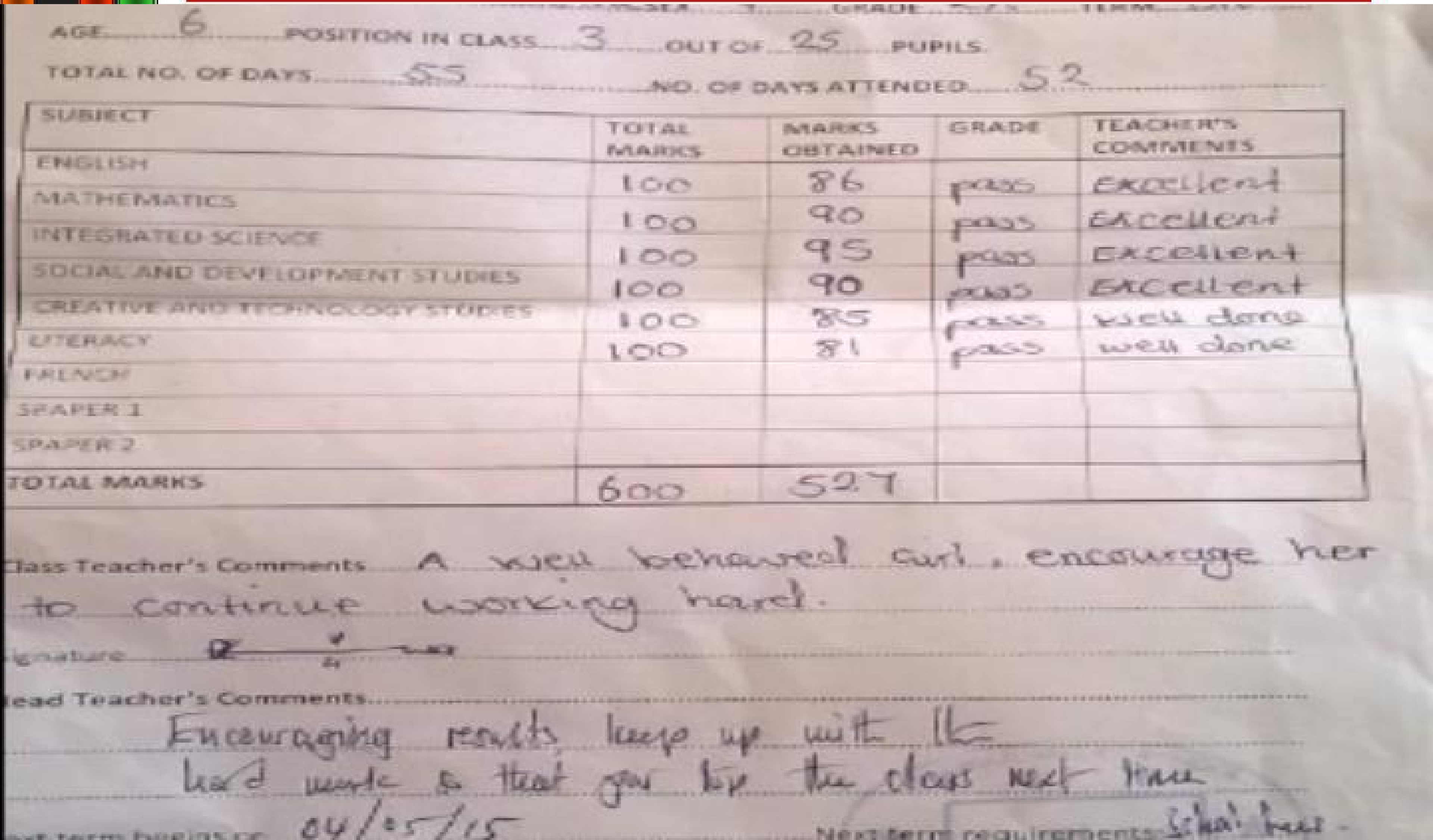
Program	Number of New In-Service Positions
Master of Medicine	9
Master of Nursing	10
BSc Nursing	35
Environmental Health	20
Midwifery	50
Theater Nursing	10
Clinical Officer – Anesthesia	10
Master of Public Health	15







What is Information?



AGE 6 POSITION IN CLASS 3 OUT OF 25 PUPILS

TOTAL NO. OF DAYS 55 NO. OF DAYS ATTENDED 52

SUBJECT	TOTAL MARKS	MARKS OBTAINED	GRADE	TEACHER'S COMMENTS
ENGLISH	100	86	pass	Excellent
MATHEMATICS	100	90	pass	Excellent
INTEGRATED SCIENCE	100	95	pass	Excellent
SOCIAL AND DEVELOPMENT STUDIES	100	90	pass	Excellent
CREATIVE AND TECHNOLOGY STUDIES	100	85	pass	well done
LITERACY	100	81	pass	well done
FRENCH				
SPAPER 1				
SPAPER 2				
TOTAL MARKS	600	527		

Class Teacher's Comments A well behaved girl, encourage her to continue working hard.

Signature

Head Teacher's Comments Encouraging results keep up with the hard work so that you top the class next time

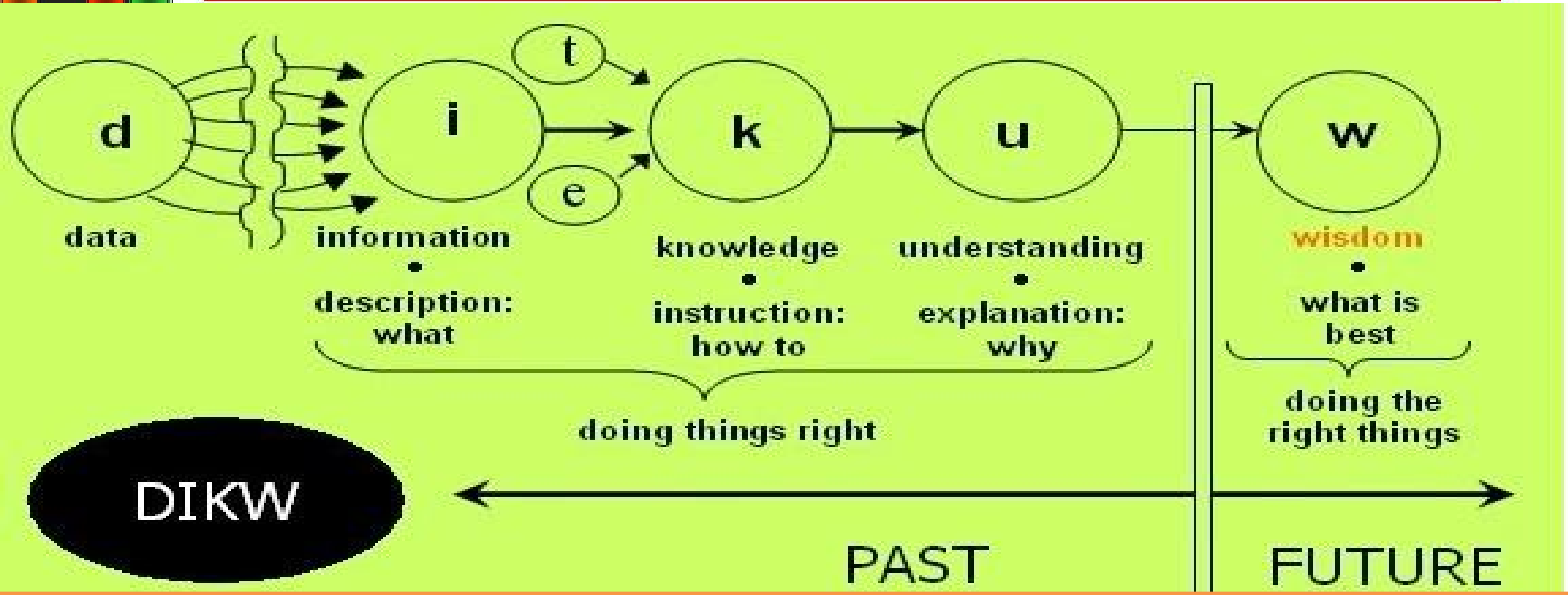
04/05/15

Next term requirements: School fees -

What is Information?

- Information is processed data that provide meaning for observations made about a phenomenon of interest.
- These are data that are processed, organized, structured or presented in a specific context.
- A school report card is a basic example of information as it provides meaning about a pupil’s performance in a class.

Data, Information, Knowledge and Wisdom Pyramid



- Data are a mere collection of symbols or signs, details or facts about a phenomenon/ event.
- Information are data endowed with context, meaning and purpose.
- Knowledge is a contextual information based on information from which patterns can be identified and be applied

Lets think of this???
DIKW

Think of the word “WifiPassword”.

•The word alone is **data**.

•**Understanding** that it is a word is **information**.

•**Knowing** it is your wifi password is knowledge.

•**Using** it to access your wireless is wisdom.

******But everything starts with data.***

Types of Data

- Basically two types of Data:

A. Numeric

- These are data expressed as numbers/counts. Can be discrete or continuous
  - **Discrete:** these are counts that take whole numbers. E.g number of children, number of students in DEM 1110 class, population of Zambia, number of people not employed.

Types of data

- Continuous- numerical data measured on an unbroken scale
  - Age: . . .18, 19, 20, 2
  - Weight: . . .150.2, 15
  - Temperature: … 37, 37.2, 37.9, 38…

Think of some others
-height -blood pressure

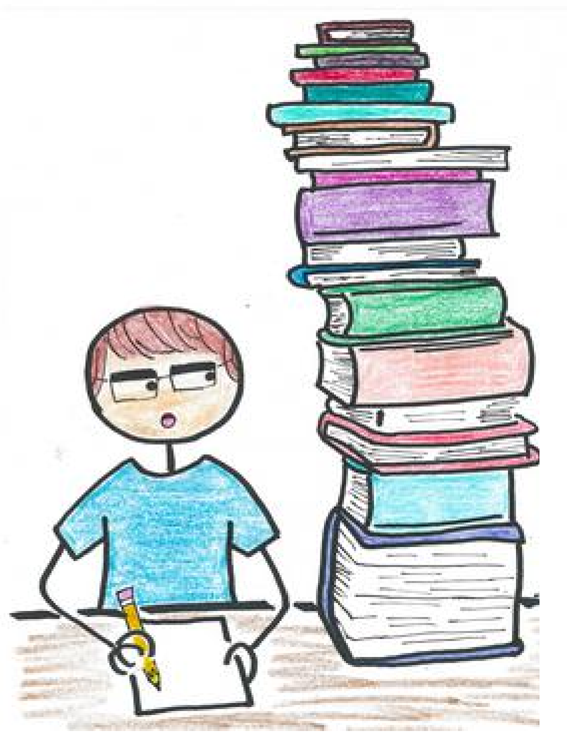
B. Categorical

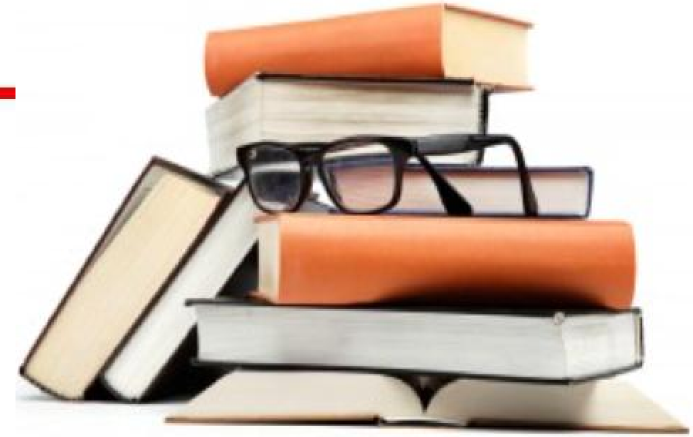
data that are divi

- Age Categories:
- Sex: Male, Female
- HIV status: positive, negative

Think of some others
-Marital Status (Married, single) -Trimester (1, 2, 3)

Recap.....





Recap!

- What is the difference between data and Information
- Why do we need information/Data
- What are the two types of data
  - Numeric
    - Discrete
    - Continuous
  - Categorical

Statistics

- Statistics is a mathematical science involving the **collection, interpretation, measurement, enumerations or estimation analysis, and presentation of natural or social phenomena**, through application of various tools and technique the **raw data becomes meaningful and generates the information's for decision making purpose**.

Statistics

- **Statistical methods** are generally concerned with the study of how to **Collect, Organize, Analysis, Present and Interpret numerical information from data.**
- **Statistics:** Statistics are a numerical value that describes some property of a phenomenon/event.
- These are numbers with context which may be social, economic or demographic.

Statistics

- Statistical methods are required to ensure that data are interpreted correctly and that apparent relationships are meaningful.

- The processes of transforming data into information is called **Data Analysis.**

Types of Statistics

- **Descriptive statistics –** Involves methods of organizing, and summarizing information from data
  - Descriptive statistics are used to describe the basic features of the data
  - simply describes what is or what the data shows
  - Descriptive statistics help us to simplify large amounts of data in a sensible way
- **Inferential statistics** - Involves methods of using information from a **sample** to draw conclusion about the **population**.

Statistic and Parameter

- **Statistics:**
  - Characteristics of a Sample, used to estimate characteristic of a population in absence of measure about the entire population
- **Parameter:**
  - Characteristic of a population, this is a real value when the entire population is measured/enumerated

Nature of Social Economic Statistics

- **Economic statistics:** These are a historical record of economic activity, capable of guiding the understanding of an economic system and guiding the formulation of policy within the system.
- Quantitative information on **manpower, production, distribution, transport, foreign trade, prices, employment, investments, national income and expenditures** are examples of economic statistics.

Nature of Social Economic Statistics

- **Social statistics:** This refers to data generated on the condition and quality of life of the people.
- Statistical information on:
  - Housing
  - Education,
  - Health, public safety
  - Access to water, etc

Common Concepts in Statistics

- **Level:** The amount of something. How much of a phenomenon/Event of concern exists at a particular time. This can be social, economic or demographic e.g.
  - # of cholera cases
  - # of people with access to banking services
  - # of UNZA momas with Brazilian hair

Common Concepts in Statistics

- **Pattern:** This is the regular way in which an even happens or unfolds
  - A pattern follows a rule in which its is arranged or presented
  - E.G. 2, 7, 12, 17, 22
- **Trend:** This is the general direction in which an event or phenomenon is changing or manifests itself over a given period of time (how has been mortality rate).

Uses of Social Economic Statistics

- Planning for national development
- Provision of services (Education, Health, Housing, water, Roads, Banking etc)
- Study human behavior
- Holds a central in research; Industry, Commerce, Trade etc
- Construction of systems of national accounts
- Construction of Economic Models
- Policy formulation and decision making

Social Economic Statistics

Academic or student

University students and faculty, institute members, and independent researchers

Corporate, government, or NGO researcher

Technology or product developers, R&D specialists, and government or NGO employees in scientific roles

Medical

Health care professionals, including clinical researchers

Problems of collecting Statistics in Zambia

- Conceptual problem
- Problem in the statistical system
  - Shortage of well-qualified statistical manpower
  - Inadequate coordination, cooperation and collaboration among major producers of statistics in the country
  - Inadequate funding of the statistical agencies
  - Administrative bureaucracy and red tapism

Problems of collecting Statistics in Zambia

- Problem in Society
  - Lack of statistical awareness
  - Illiteracy/ innumeracy
  - Cultural /religious problems
  - Language problem
  - Poor social facilities

Major Producers of Statistics in Zambia

A. Central Statistical Office (CSO)

- Main statistical agency responsible for developing and management of statistics
- Established by an act of parliament, Cap 127 of the laws of Zambia
- Authority and custodian of all official statistics
- Its under the Ministry of National Development Planning

Subject Mater areas under CSO

- Demography
- Economic Statistics
- Social statistics
- Agriculture statistics
- Labor statistics
- Information , Research and Dissemination

Functions of the Central statistical Office

- To coordinate the national statistical system
- To advise government, ministries and other agencies on all matters related to statistical development
- To develop and promote use of statistical standards and appropriate methodologies in the system
- To collect, compile, analyze, interpret and disseminate information alone or in collaboration with other agencies, both government and non-governmental agencies
- To develop and maintain a comprehensive national data bank by encouraging unit of line ministries and agencies develop their sectoral data bank and forward to CSO

Functions of the Central statistical Office

- To provide a focal point of contact with international agencies on statistical matters
- To carry out all other functions relating to statistics as might be assigned by government

Statistical Activities of CSO

- Conduct the National Census
- Conduct the Demographic and Health surveys (ZDHS)
- Living Conditions and Monitoring survey (LCMS)
- Economic Census (EC)
- Agriculture Survey
- Labor force survey
- Zambia HIV/AIDS Impact Assessment (ZAMPHIA)
- Monthly Food Baskets

Major Producers of Statistics in Zambia

B. Central Bank of Zambia (Banks of Zambia)

- The Bank of Zambia publishes banking, financial, foreign trade statistics etc through its;
- Annual Report and Statement of Account
- Economic and Financial Review
- Monthly Report
- Principal economic and financial indicators
- Statistical Bulletins, e.t.c

Major Producers of Statistics in Zambia

C. **Other Sources**

•**Line Ministries and government agencies (ZDA, ZICTA, RDA, and other Ministries)**

•Bilateral organizations (World Bank, UN, WHO etc)

•UN Department of Economic and Social Affairs, Population Division.

http://www.un.org/esa/population/unpop.htm

•UN Demographic year book

•UNDATA

•Web bases sources (US. Census Bureau, UN stats, etc)

• World Health Organization (WHO).

http://www.who.int/en.

- Human Mortality Database.

http://www.mortality.org.

- United Nations Population Fund

http://www.unfpa.org/public/

OECD

The Organisation for Economic Co-operation and Development

http://www.oecd.org/home/0,2987,en_2649_201185_1_1_1_1_1,00.html\

Data Collection Types

Data Collection Types

- Data is an important and the driver of any program, project or policy implementation.

- It is therefore important to understand:
  - what are the sources of data?
  - How data are collected?
  - What type of data programs, projects or policies generates?
  - Components of data collected,
  - What methods of data collection are used?

Data Collection Types

Data Collection entails;

- A process of preparing and collecting data
- A systematic gathering of data for a particular purpose from various sources.
  - Data are the basic inputs to any decision making;
  - The raw material for socio economic investigation and analysis is data;
  - The type of data collected for a given purpose may be from **Primary** or **Secondary** sources

Primary Data Collection

- This is a data collection approach where data is collected at first hand for a specific purpose.
- The basic sources of primary data are:

-Interviews, observation and administrative processes such as medical examinations, voter registration, national registration cards Census

Primary Data Collection

Primary data are collected where:

- The needed information does not exist elsewhere
- The needed information exist but is not **reliable/ validity** can not be assured
- Collecting the data at first hand is only way such information can be obtained

Primary Data Collection

Advantages of Primary Data

- Targeted issues are addressed
- Data interpretation is better
- Proprietary issues
- Address specific issues
- Greater control

Primary Data Collection

Disadvantages

- Costly
- Time consuming
- Inaccurate feed-back
- More resources are required
- Differences in conceptual definitions

Secondary Data Collection

Secondary Data Collection

- These are data which already exist and may be adapted for use in the current survey.
- Such data are collected originally for another purpose.
- Secondary data can be sourced from:
  - publications and records of governments
  - non-government organizations, journals of universities and research institutes,
  - Media organization
  - administrative records.

Secondary Data Collection

- secondary data collection to be effective and useful with reasonable degree of confidence, the data must be assessed. This involves checking for the following;
  - The source of the data
  - The purpose of which it was collected
  - The method of data collection used
  - Definition of terms used
  - Coverage and changes overtime, if any
  - Method of analysis

Secondary Data Collection

Advantage	**Disadvantages**
• Faster • Less Expensive • Less activities (Field trip, Survey etc.)	• Not easily available • Not adequate • May not meet the needs of a researcher/data user • Outdated information • Variation in definition • Inaccurate or bias

Routine vs. Non-routine Data CollectionTypes

- Routine data
  - Collected continuously (daily, patient by patient, monthly, etc.)
  - Collection is on an ongoing basis e.g. hospital patient data, registration for NRC
  - Data collected at the point of service
- Non-Routine (periodic) data
  - Collected at certain periods of time, or over a specific period of time
  - Come from special studies or surveys carried out for specific purposes (E.G. ZDHS, ZAMPHIA)

Routine Data – Sources

- Data collection/Entry Instruments
  - Registers
  - Daily activity sheets
  - Patient cards
  - Program-specific forms
  - EHS (SmartCare)

Strengths & Limitations of Routinely Collected Data

- Strengths
  - Often low cost since data is routinely collected
  - Usually collected at the point of service delivery (i.e., health facility) then obtained at district, provincial and national levels (to offer 'complete picture' of the country)
  - Data collection often coincides with program implementation therefore these data can be used to monitor and evaluate program goals
  - Can help identify abnormal trends in the occurrence of events

- Limitations
  - Time-consuming, can be burdensome for health staff
  - Data quality not always ensured

Example of Non-Routine Data Sources

In addition to routinely collected health data, Zambia has periodic surveys such as:

- Demographic and Health Survey (DHS)
- Antenatal Clinic Sentinel Surveillance (ANCSS)
- Sexual Behaviour Survey (SBS)
- National Census
- MDG's Survey (planned)

Strengths and Limitations of Non-Routine Data

- Strengths:
  - Surveys especially useful if other data are not available or inadequate
  - Surveys can be tailored to fit specific measurement objectives
  - Surveys yield cost-efficient data on population and services useful for facilities, providers and clients

- Limitations:
  - Surveys are expensive and time consuming
  - Difficulty recall (DHS costs around $9 million and takes many months to complete)
  - Survey sampling design and analysis may be complex

SOURCES OF SOCIAL AND ECONOMIC STATISTICS

A. CENSUS

- This is the total process of collecting, compiling and publishing demographic, economic and social data pertaining at a specified time to all persons in a given country or delimited territory.
- **Population Census** provides the fullest and most reliable picture of the country's population and its characteristics at the "Census Day " (a particular point in time to which the census relates)

Characteristics of a Good Census

a. **Universality**- every unit or individual must be counted
b. **International comparability-** Should be conducted using the standards and principles of all census worldwide
c. **Sponsorship**- Because it is an expensive venture
d. Must **focus on an individual** (information collected pertains to each and every individual)
e. **Periodicity**- it must have a specific period of time e.g 5 or 10 years
f. **Simultaneity**- it should be taking place at the same time with other countries for the purpose of comparisons.
g. **Legality** – it must have a legal basis. E.g the CSO is mandated by an act of parliament to conduct a census.

Types of Census

Two types of Population Census:

A. **De-facto Population Census;**

- This type of census involves counting of only those who are **physically** present during the census
- This counts people from where they are found (People who spent a night at a household) regardless of whether they stay there or not

Defacto Population Census

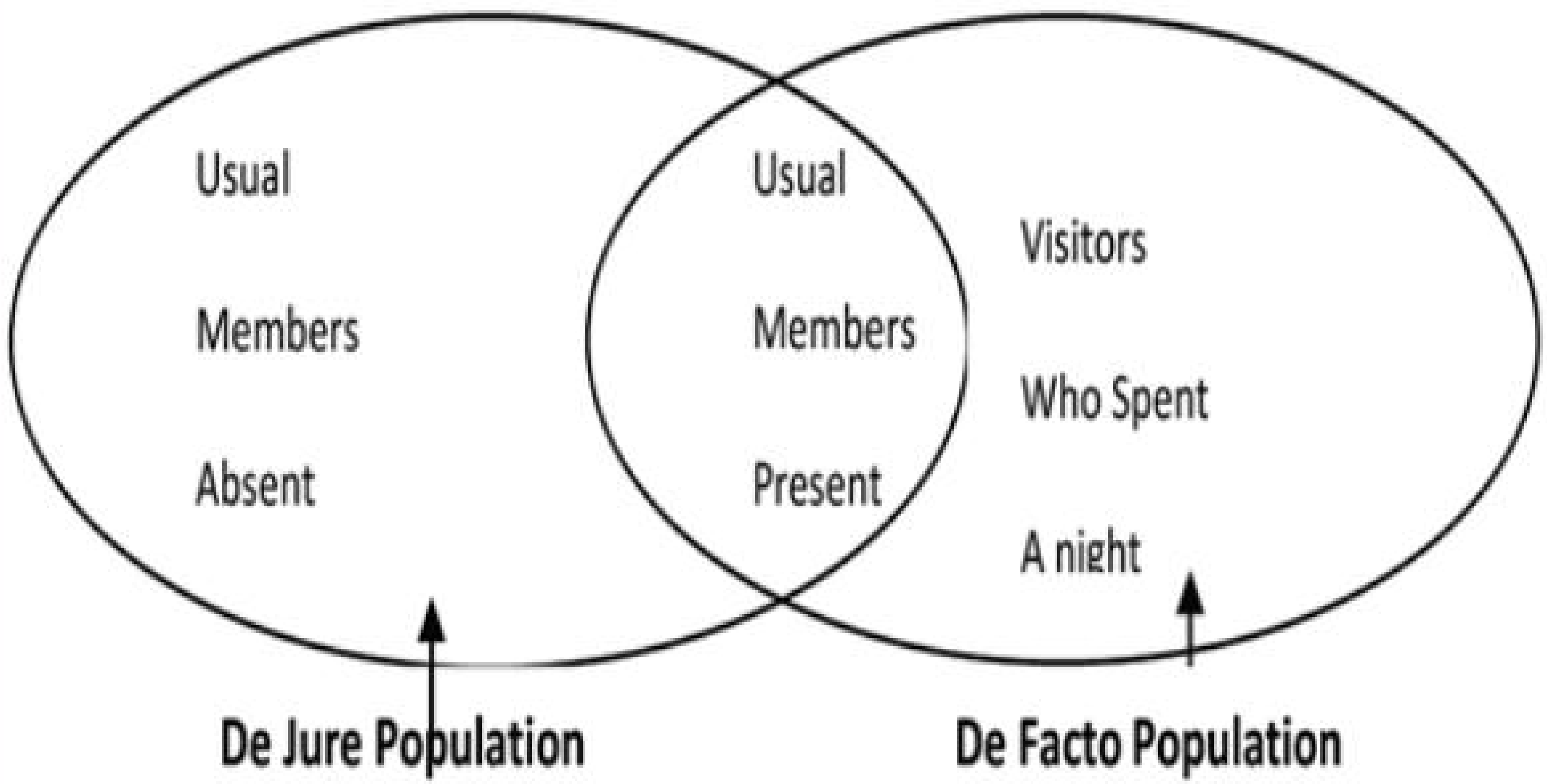
- **Defacto Population Census excludes;**
  - Foreign diplomatic personal accredited to Zambia
  - Zambian nationals accredited to foreign embassies and their family members who live with them abroad
  - Zambian migrant workers and students in foreign countries who might not be in the country at the time of the census

Types of Census

B. **De-jure Population Census;**

- Involves counting people who have been permanent residents of a specific area
- Does not matter whether a person is present or not
- Counts the usual household members present and usual household members temporarily absent at the time of the census
- In de-jure methods, populations in institutions such as hospitals, universities/colleges, prisons, schools are counted as members of their usual household

De-jure and De-facto Census Population



De-jure and De-facto Census Population

- De jure population is the true resident population of a country
- Its used as denominator for calculation of vital indicators (sex ratio, Births, deaths)
- Does not include the social, economic characteristics of people who absent from the household at the time of the census
- Zambia uses the Dejure population count (Usual members absent)

Importance of a Census

1. **To know the population size**; Population size living in a country, structure and its composition
2. **Determination of revenue estimates;** to know the size of the taxable adult population and estimate amount of revenue expected from the sector
3. **Forecast future social and economic needs**: education, health, housing, food

Importance of a Census

4. Determines the level of unemployment
5. Number of immigrants
6. Equitable distribution of resources
7. Provision of social amenities
8. Delimitation of districts, constituencies
9. Determination of the level of working age, man power
10. Provision of social amenities
11. Investment decision

Problems Associated with Population Census

1. High levels of illiteracy: because of this it becomes difficult to get accurate, relevant and useful statistics
2. Political barriers; since the census is used in the delimitation of areas, allocation of resources.
3. High Cost: A lot of money is required to conduct the census, some countries cannot afford a census
4. Geographical barriers
5. Lack of adequate trained manpower
6. Religious reasons
7. Geographical re-demarcations (delimitation of new districts)
8. incompleteness and Omissions (areas and Individuals) (PES)

History of Census Taking in Zambia

Pre Independence:

- First census was conducted in 1911, only covered non Africans
- Other census that followed were in 1921, 1931, 1946, 1951, 1956 and 1961 (all covered non Africans
- Estimates of Africans were based on tax registers
- First attempt to count Africans was through the Demographic survey conducted in 1950

History of Census Taking in Zambia

- Just before Independence, there was need for a complete Census thus the first Census which included Africans in 1963.

History of Census Taking in Zambia

Post Independence:

•So far Five Censuses have been conducted in;

- 1969 (4, 661, 331),
- 1980 (5,661, 539),
- 1990 (7, 759, 117),
- 2000 (9,885, 591)
- 2010 (13, 092, 666).

http://www.ined.fr/en/everything_about_population/videos/population-growth-today/

Census Micro Data

- Zambia undertakes Traditional Censuses, that is **personal visits by Enumerators to households**
- The scope of work is normally 100 percent Collects both De Jure and De Facto population
- A **key respondent** is identified in each household
- Period of collection is on average 1 month
- Forms are brought to Centre within a month for completion

Contents of Census Micro Data

The Census Data includes:

- Identification variables (Province, District etc)
- Household structure (membership status)
- Demographic and social variables
- International Migration
- Education
- Economic Activities
- Fertility and Mortality
- Civil Registration
- Household and Agriculture variables

REPUBLIC OF ZAMBIA

2010 CENSUS OF POPULATION AND HOUSING

CENTRAL STATISTICAL OFFICE

IDENTIFICATION

Province	1 2 3 4 5 6 7 8 9	Constituency	0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9
District	0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9	Ward	0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9
		Region	Rural 1 Urban 2
CSA No.	0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9	Housing Unit No. (HUN)	0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9
SEA No.	0 1 2 3 4 5 6 7 8 9	Household No. (HHN)	0 1 2 3 4 5 6 7 8 9
Census Building No. (CBN)	0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9	Village/ Locality Name	
Residential Address		Chief's Area	0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9

INTERVIEW STATUS

- Completed (occupied) 1
- Non-contact (occupied) 2
- Not interviewed (vacant) 3
- Non residential 4
- Refused 5
- Other 6

MARK HERE IF MORE THAN ONE QUESTIONNAIRE 1

Questionnaire ☐☐

of ☐☐

SECTION P - QUESTIONS ON POPULATION FOR ALL MEMBERS OF THE HOUSEHOLD

P1 NAMES of usual residents and visitors	P2 Membership status? Usual member Present / Usual member Absent / Visitor	P3 What is (NAME'S) relationship to the head of household? Head of Household / Spouse / Own Son/Daughter / Step Son/Daughter / Parent / Brother/Sister / Nephew/Niece / Son/Daughter-in-Law / Grandchild / Parent-in-Law / Cousin / Other Relative / Unrelated	P4 Is (NAME) male or female? Male / Female	P5 How old was (NAME) at his/her last birthday? *If less than 1 year enter "00"* *In Years*	P6 PLACE OF BIRTH Where was (NAME) born? *Write District/Country name then code*	P7 Was this part of the district rural or urban at the time of birth? Rural / Urban / Outside Zambia	Is (NAME) Zambian? P8 If Zambian, code here then skip to P11	P9 If non-Zambian, code here then write name of country	P10 What is the main purpose of your stay in Zambia? Employment / Family formation/ reunification / Education/training / Settlement / Refuge/Asylum / Investor / Tourist / Other	P11 What is (NAME'S) religion? Catholic / Protestant / Muslim / Hindu / Buddist / Bahai faith / Other / None
1	1 2 3	1 2 3 4 5 6 7 8 9 10 11 12 13	1 2	☐☐	.... ☐☐☐	1 2 3	1	2 .... ☐☐☐	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8
2	1 2 3	1 2 3 4 5 6 7 8 9 10 11 12 13	1 2	☐☐	.... ☐☐☐	1 2 3	1	2 .... ☐☐☐	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8
3	1 2 3	1 2 3 4 5 6 7 8 9 10 11 12 13	1 2	☐☐	.... ☐☐☐	1 2 3	1	2 .... ☐☐☐	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8
4	1 2 3	1 2 3 4 5 6 7 8 9 10 11 12 13	1 2	☐☐	.... ☐☐☐	1 2 3	1	2 .... ☐☐☐	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8
5	1 2 3	1 2 3 4 5 6 7 8 9 10 11 12 13	1 2	☐☐	.... ☐☐☐	1 2 3	1	2 .... ☐☐☐	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8
6	1 2 3	1 2 3 4 5 6 7 8 9 10 11 12 13	1 2			1 2 3	1	2	1 2 3 4 5 6 7 8	1 2 3 4 5 6 7 8

B. SURVEYS

- Sample surveys are among the major sources of socio-economic and demographic data in Zambia.
- Sample surveys acknowledge the importance of censuses. However, censuses are conducted at large intervals e,g ten years and they tend to collect information with less detail or depth.
- A sample survey is a scientific data collection process where only a proportion of the population is involved.

Approaches of Capturing Information in Surveys

Two Basic approaches:

•**Retrospective:** Focuses on events that have already occurred/happened. Usually data is collected subject to 12 months prior the census (e.g. did you use any contraception in the last 12 month, did you have fever in the last 2 weeks)

•**Prospective:** Focuses on events as they are happening. E,g participating in class, the spread of ebola virus etc

Advantages of Sample Surveys

- Provides a cheaper alternative for timely data and convenient social, economic and demographic data pertaining to conditions under which people live, their wellbeing, activities in which they engage in such as demographic, social, cultural and economic factors which influence behavior as well as socio-economic changes.
- Provides a more flexible method of data collection, making it an excellent choice for meeting data users' needs.

Advantages of Surveys

- Collects data at a greater depth/detail due to larger time given for interviews and smaller interview workloads **(detailed and specific subject matter).**
- In reality not all the data needs of a country can be met through census undertaking. provide a mechanism for meeting the additional and emerging needs on a continuous basis.
- Important to capture Social, economic and demographic characteristics that change frequently,
- Easy to observe events over time in a survey than a census

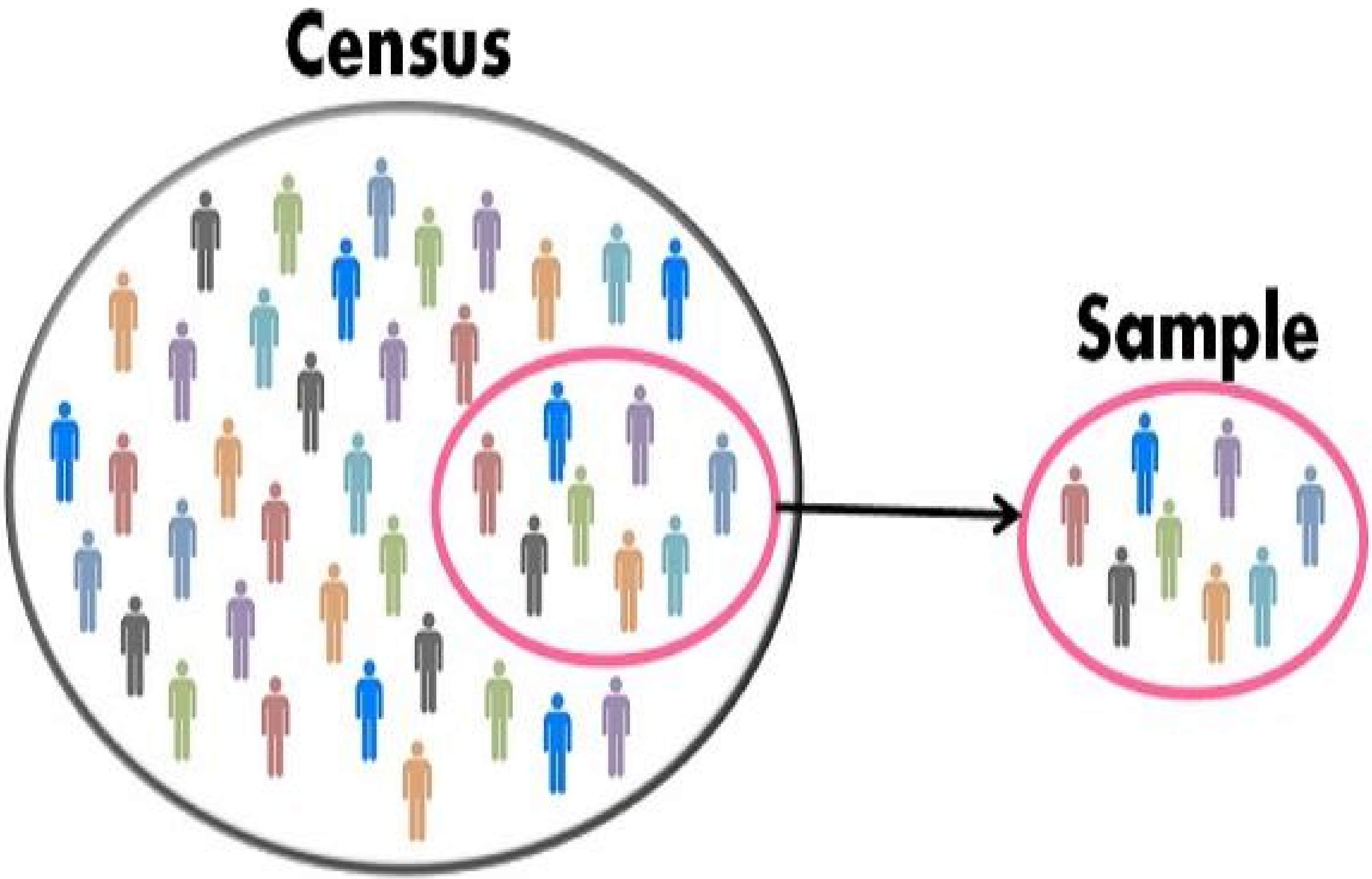
Disadvantages of Surveys

- Prone to sampling errors
- Prone to abuse
- Poor response rate because they are not legally bidding as the census
- The size of the sample only ensures that reliable statistics can be shown to a very limited geographic detail (if not properly done)
- Coverage is only extended to a limited geographical area

Link between Census and Surveys

- Census data/statistics serve as benchmark for analyzing and evaluating survey data
- Census is often used as a sampling frame for selecting the population to be surveyed in a census (means of selecting a specific survey population group)
- Surveys have been used to determine the amount of error in a census (Post enumeration survey)

Difference between Census and survey



BASIS FOR COMPARISON	CENSUS	SAMPLING
Meaning	A systematic method that collects and records the data about the members of the population is called Census.	Sampling refers to a portion of the population selected to represent the entire group, in all its characteristics.
Enumeration	Complete	Partial
Study of	Each and every unit of the population.	Only a handful of units of the population.
Time required	It is a time consuming process.	It is a fast process.
Cost	Expensive method	Economical method
Results	Reliable and accurate	Less reliable and accurate, due to the margin of error in the data collected.
Error	Not present.	Depends on the size of the population
Appropriate for	Population of heterogeneous nature.	Population of homogeneous nature.

C. Vital Registration System

- The vital registration system involves the continuous and permanent recording of vital events pertaining to **births, deaths, marriages, divorces, legal separations, annulments etc.**
- In developing countries the system is still under problems
- The information is collected by the civic authorities such as the city, district, and municipal councils.
- In Zambia National Registration office (Ministry of Home Affairs) is responsible for registration of vital events
- The vital registration system gives **certificates** to the occupancy of events such as **Birth, death and marriage certificates.**

Uses of Vital Events Certificates

a. **Birth Certificates (issued by the local authority upon child's birth/ National Registration Office)**

- Prove of identify
- Claiming benefits (Inheritance) (in case of parents dying)
- Verifying Childs age
- Verifying parentage and relationship of the child

Uses of Vital Events Certificates

b. Death Certificates (issued by the attending doctor and the burial permit is issued by the civic authority)

- Used in pension schemes to prove an individual's death so that the surviving relatives can claim benefits
- Used to transport bodies from one place to another for burial
- Used by social welfare authorities
- Used to estimates birth rates

Uses of Vital Events Certificates

c. Marriage Certificates (issued by the local authorities and other institutions)

- Proof of marriage
- Claiming of benefits when spouse dies, divorces
- Used to observe marriage and fertility patterns

Types of Registration of Vital Events

Broadly two types of vital events Registration:

a.Passive System:

- This approach relies on the good will and cooperation of the public (individuals) to report events as they occur.
- This implies that registration personnel wait until informants (public/individuals) report the events)

Types of Registration of Vital Events

b. Active System

- Under this approach, the registration institution employs registration personnel who go round to canvass the vital events that occurred rather than wait for informants to do so.
- As part of the system, chiefs, teachers, health workers, midwives, religious leaders and other local government functionalities could be used as notifies

Problems of VRS

- Underfinance from the part of government (lack of funds) to support the recording of vital events. Reliance has been placed on individuals to report and have vital events recorded
- It's not mandatory as such people do not see the utility of registering vital events such as births, deaths etc making the whole registration of vital events fragile
- Due to high Illiteracy levels, people do not see the need of registering (especially in rural areas

Problems of VRS in Zambia

- Lack of infrastructure (both systems and physical infrastructure) to support the registration of vital events
- Lack of administrative structures at lower levels (ward, constituency)
- Low status and lack of expertise
- The time lapse between the occurrence of a vital event and registration make most vital statistics unreliable and fraud
- Problems of rural areas

Improving Vital Registration System For National Economic Planning And Social Development

- Strengthening the legislation system governing the registration of vital events
- Making the registration of vital events compulsory and continuous
- Strengthening of infrastructure through local government structures to the lowest organ of administration
- Strengthen advocacy on registration of vital events

Current Strategies to Capture Vital

- Sample Vital Registration System (SAVY) (Livingstone- Birth and death Registration)
- Registration of death in Lusaka through data extraction from burial permits and certification of death from hospitals
- Strengthening the ehealth system in health facilities

Current Strategies to Capture Vital

D. Population Register

- The Population Register is a comprehensive identity database that is maintained on a continuous basis and contains all people in a country and their characteristics.

Such as;

- Date of birth
- Occurrence of any event such marriage, divorce, place of occurrence, characteristics of siblings or children and that of the deceased
- Characteristics of the parents of the new born
- Cause of death in-case of the deceased
- Migrations (in and out)

Population Register

- If a population register is well maintained, it gives all the details and they might not be need for a census
- Some countries such as Austria, Denmark, Finland, Israel, Japan, The Netherlands, Norway and Sweden have removed the need for censuses by adopting a system whereby population statistics are continuously updated based on information recorded in **administrative registers (Population Register)**.

Population Register

- Population registers are designed in such a way that in each locality e.g the council, city or town there will be a listing process (register) where all the characteristic of people are entered.
- Each person is assigned a unique identifying number which enables information across different registers (i.e. birth, death, marriage etc.) to be linked and for central records to be updated.
- The linking of records means the registers can also be used for longitudinal data which follows an individual across the life course.

Population Register Procedure

The procedure is very simple but sitting on a very complex system;

- When someone is born a card is opened and the information is updated as the person progresses in life (given a unique identifier)
- When one moves, he/she is registered as a migrant and the both the place of departure and arrival are notified
- When one dies, the card is closed to indicate the terminal end of the person life.

Basic Data in Population Register

- The basic data recorded by population registers are births, deaths, marriages, nationality, migration etc.
- Population registers may also cover family relationships (i.e. parents, spouses) and dwellings, allowing data on individuals to be linked by family and household.
- Some Nordic countries also include information on health, educational attainment, employment and income within the population register.
- Population register data can be used to trace family genealogy

Example. Dutch Genealogy

- full name
- birth place
- birth date
- death date (if death occurred within that particular 10-year period)
- marital status
- marriage date (if marriage occurred within that particular 10-year period)
- relationship to head of household
- religion

Example. Dutch Genealogy

- sometimes: tax class
- date of registration
- date of moving to that address (if within that particular 10-year period)
- place or address from where he came
- date of moving from that address (if within that particular 10-year period)
- place or address to where he moved
- notes

Zambia’s History of Population Register

- In Zambia, they were what were called Village Registers which had all the characteristics of the people in the village
- The Village Registers where deposited at the Chief’s palace.
- When someone is born, information was recorded; when one moves he/she was given a letter addressed to the place of destination.

Advantages of Population Registers

- The continuous nature of data collection for population registers means that the information available is current.
- Population registers can be updated almost instantaneously, meaning that there is no long waiting period for the data to be collated and analyzed.
- There is a potential for linked records, meaning that (anonymized) individuals can be followed through the life course for research purposes

Disadvantages of Population Registers

- Very expensive to maintain as it requires setting up of structures
- Its passive in nature
- Does not work effectively in countries with high illiterate population
- The type of data available is determined by the nature of the administrative departments collecting it.
- Events occurring before the start of registration, or occurring while the individual was abroad, may be excluded.
- The fact that the record is permanent, follows through life, and is linked between departments, may be unacceptable to some with concerns of a “Big Brother” style society. There are also concerns regarding the potential for data leaks.